Network & Protocols, Middleware UI, Application Logic

System Design(SD)(22CS024)

Bachelor of Engineering- Computer Science & Engineering

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Network Protocol

Network protocols are a set of rules that are responsible for the communication of data between various devices in the network. These protocols define guidelines and conventions for transmitting and receiving data, ensuring efficient and reliable data communication.

What is Network Protocol?

A network protocol is a set of rules that govern data communication between different devices in the network. It determines what is being communicated, how it is being communicated, and when it is being communicated. It permits connected devices to communicate with each other, irrespective of internal and structural differences.

How do Network Protocols Work?

- It is essential to understand how devices communicate over a network by recognizing network protocols. The Open Systems Interconnection (OSI), the most widely used model, illustrates how computer systems interact with one another over a network.

- The communication mechanism between two network devices is shown by seven different layers in the OSI model. Every layer in the OSI model works based on different network protocols. At every layer, one or more protocols are there for network communication.

- To enable network-to-network connections, the Internet Protocol (IP), for instance, routes data by controlling information like the source and destination addresses of data packets. It is known as a network layer protocol.

Types of Network Protocols

- In most cases, communication across a network like the **Internet** uses the **OSI model.**
- The OSI model has a total of seven layers.
- Secured connections, network management, and **network communication** are the three main tasks that the **network protocol** performs.
- The purpose of protocols is to link different devices.
- The protocols can be broadly classified into three major categories:
- **Network Communication**
- **Network Management**
- **Network Security**

1. Network Communication

Communication protocols are really important for the functioning of a network. They are so crucial that it is not possible to have computer networks without them. These protocols formally set out the rules and formats through which data is transferred. These protocols handle syntax, semantics, error detection, synchronization, and authentication. Below mentioned are some network communication protocol.

Hypertext Transfer Protocol(HTTP)

It is a layer 7 protocol that is designed for transferring a hypertext between two or more systems. HTTP works on a client-server model, most of the data sharing over the web is done through using HTTP.

- **Transmission Control Protocol(TCP)**
- TCP layouts a reliable stream delivery by using sequenced acknowledgment. It is a connection-oriented protocol i.e., it establishes a connection between applications before sending any data. It is used for communicating over a network. It has many applications such as emails, FTP, streaming media, etc.

- **User Datagram Protocol(UDP)**
- It is a connectionless protocol that lay-out a basic but unreliable message service. It adds no flow control, reliability, or error-recovery functions. UPD is functional in cases where reliability is not required. It is used when we want faster transmission, for multicasting and broadcasting connections, etc.

- **Border Gateway Protocol(BGP)**
- BGP is a routing protocol that controls how packets pass through the router in an independent system one or more networks run by a single organization and connect to different networks. It connects the endpoints of a LAN with other LANs and it also connects endpoints in different LANs to one another.

- **Address Resolution Protocol(ARP)**
- ARP is a protocol that helps in mapping logical addresses to the physical addresses acknowledged in a local network. For mapping and maintaining a correlation between these logical and physical addresses a table known as ARP cache is used.

- **Internet Protocol(IP)**
- It is a protocol through which data is sent from one host to another over the internet. It is used for addressing and routing data packets so that they can reach their destination.

• **Dynamic Host Configuration Protocol(DHCP)**

•it's a protocol for network management and it's used for the method of automating the process of configuring devices on IP networks. A DHCP server automatically assigns an IP address and various other configurational changes to devices on a network so they can communicate with other IP networks. it also allows devices to use various services such as NTP, DNS, or any other protocol based on TCP or UDP.

•**2. Network Management**

•These protocols assist in describing the procedures and policies that are used in monitoring, maintaining, and managing the computer network. These protocols also help in communicating these requirements across the network to ensure stable communication. Network management protocols can also be used for troubleshooting connections between a host and a client.

- **Internet Control Message Protocol(ICMP)**
- It is a layer 3 protocol that is used by network devices to forward operational information and error messages. ICMP is used for reporting congestions, network errors, diagnostic purposes, and timeouts.

- **Simple Network Management Protocol(SNMP)**
- It is a layer 7 protocol that is used for managing nodes on an IP network. There are three main components in the SNMP protocol i.e., SNMP agent, SNMP manager, and managed device. SNMP agent has the local knowledge of management details, it translates those details into a form that is compatible with the SNMP manager. The manager presents data acquired from SNMP agents, thus helping in monitoring network glitches, and network performance, and troubleshooting them.

Gopher

It is a type of file retrieval protocol that provides downloadable files with some description for easy management, retrieving, and searching of files. All the files are arranged on a remote computer in a stratified manner. Gopher is an old protocol and it is not much used nowadays.

File Transfer Protocol(FTP)

FTP is a Client/server protocol that is used for moving files to or from a host computer, it allows users to download files, programs, web pages, and other things that are available on other services.

- **Post Office Protocol(POP3)**
- It is a protocol that a local mail client uses to get email messages from a remote email server over a TCP/IP connection.
- Email servers hosted by ISPs also use the POP3 protocol to hold and receive emails intended for their users.
- Eventually, these users will use email client software to look at their mailbox on the remote server and to download their emails.
- After the email client downloads the emails, they are generally deleted from the servers.

- **Telnet**
- It is a protocol that allows the user to connect to a remote computer program and to use it i.e., it is designed for remote connectivity.
- Telnet creates a connection between a host machine and a remote endpoint to enable a remote session.

3. Network Security

- These protocols secure the data in passage over a network. These protocols also determine how the network secures data from any unauthorized attempts to extract or review data. These protocols make sure that no unauthorized devices, users, or services can access the network data. Primarily, these protocols depend on encryption to secure data.

- **Secure Socket Layer(SSL)**
- It is a network security protocol mainly used for protecting sensitive data and securing internet connections. SSL allows both server-to-server and client-to-server communication. All the data transferred through SSL is encrypted thus stopping any unauthorized person from accessing it.

- **Hypertext Transfer Protocol(HTTPS)**
- It is the secured version of HTTP. this protocol ensures secure communication between two computers where one sends the request through the browser and the other fetches the data from the web server.

- **Transport Layer Security(TLS)**
- It is a security protocol designed for data security and privacy over the internet, its functionality is encryption, checking the integrity of data i.e., whether it has been tampered with or not, and authentication. It is generally used for encrypted communication between servers and web apps, like a web browser loading a website, it can also be used for encryption of messages, emails, and VoIP.

Some Other Protocols

- **Internet Message Access Protocol (IMAP)**
- ICMP protocol is used to retrieve message from the mail server. By using ICMP mail user can view and manage mails on his system.
- **Session Initiation Protocol (SIP)**
- SIP is used in video, voice, and messaging application. This protocol is used to initiating, Managing, Terminating the session between two users while they are communicating.
- **Real-Time Transport Protocol (RTP)**
- This protocol is used to forward audio, video over IP network. This protocol is used with SIP protocol to send audio, video at real-time.

- **Rout Access Protocol (RAP)**
- RAP is used in network management. It helps to user for accessing the nearest router for communication. RAP is less efficient as compared to SNMP.

- **Point To Point Tunnelling Protocol (PPTP)**
- It is used to implement VPN (Virtual Private Network). PPTP protocol append PPP frame in IP datagram for transmission through IP based network.

- **Trivial File Transfer Protocol (TFTP)**
- TFTP is the simplified version of FTP. TFTP is also used to transfer file over internet.

- **Resource Location Protocol (RLP)**
- RLP is used to assign the resource such as server, printer, or other devices over the internet to the user. It is used to locate the resource to the client for broadcast query.

Middleware: Categories, Types and Working

- Middleware are software tools that act as intermediaries between different applications, systems, or services, facilitating their communication and interaction.
- They ensure that data and requests can be exchanged smoothly and efficiently, even if the systems involved are built using different technologies.

- They handle various tasks such as data translation, message queuing, authentication, and connectivity, making it easier to integrate and manage complex software environments. The middleware serves as the link between users, data, and applications.

- Examples include **database middleware**, **web server middleware,** and **message-oriented middleware**, **Cloud Services of all kinds, enterprise application integration, and application runtimes.**

Middleware

Middleware

Purpose of Middleware

- There are several uses for middleware. It first controls access to different back-end resources. A connection pool could be established by a middleware component to enable quick and easy access to a well-liked back-end database.
- It is also capable of establishing links to topics and message queues. Moreover, middleware software may control access to cloud-based services such as Amazon Simple Storage Service.
- Middleware software is capable of executing logic according to the client's request.
- In load balancing, transaction management, and concurrent processing, middleware is crucial. Typically, middleware software can grow both vertically and horizontally to assist in distributing incoming client requests over several servers, virtual machines, or cloud availability zones.

- Securing access to back-end resources is mostly dependent on middleware
- The capacity of middleware software to challenge clients is contingent upon two requirements: a secure connection, established through technologies such as SSL, and authentication, accomplished through the use of a digital certificate or a username and password combination.

Categories of Middleware

- **Platform middleware:** It facilitates the development and delivery of software by giving application program logic a runtime hosting environment, like a container. Web servers, application servers, and content management systems (CMSs) are examples of tools that facilitate application development and delivery and are included in platform middleware.
- **Enterprise application integration middleware**: Programmers can develop business apps with the help of enterprise application integration middleware, which eliminates the need to design unique connectors for every application. By offering a layer of functionality for data integrity and B2B connection, middleware in this case facilitates software collaboration.

Types of Middleware

- **Remote Procedure Call (RPC):** A program can ask for a service from another program running on a different computer or network using this protocol, which is provided via middleware.
- **Messaging middleware:** It makes it easier for dispersed apps and services to communicate with one another.
- **Embedded middleware:** enables integration and communication between real-time operating systems and embedded applications.
- **API middleware:** lets programmers design and control the APIs in their applications.
- **Asynchronous data streaming middleware :** By duplicating data streams in an intermediary repository, asynchronous data streaming middleware facilitates data exchange between apps.
- **Transaction or transactional middleware:** This uses transaction process monitoring to make sure transactions proceed smoothly from one stage to the next.

How Does Middleware Work?

- In its most basic form, middleware allows developers to construct applications without the need to design a unique integration each time they want to link to devices, data sources, computational resources, or services.

- This is achieved by offering services that facilitate communication between various applications and services via standard messaging frameworks, including web services, XML (extensible markup language), REST (representational state transfer), JSON (JavaScript object notation), and SOAP (simple object access protocol). In most cases.

- **Incoming Request**
- A client sends a request (e.g., GET /users).
- Express routes the request through the stack of middleware.

- **Middleware Function**
- A middleware is just a function with the signature:

```
function (req, res, next) { ... }
```

- function (req, res, next) { ... }
- req: the request object (data from client).
- res: the response object (data to send back).
- next: a callback function that moves control to the next middleware.

- **Execution Flow**
- Middleware runs in the order they are defined in your app. Each middleware can:

- Process/modify request data.
- Send a response and end the cycle.
- Call next() to pass control to the next middleware.

- **Route Handlers as Middleware**
- Route functions (like app.get('/users', ...)) are also middleware, but they usually end the cycle by sending a response.

Application Logic

- Application logic in a computer network refers to the rules and processes defined by the Application Layer protocols, which allow user applications to communicate and exchange data over a network.

- These protocols handle tasks like web browsing (HTTP), email (SMTP), file transfer (FTP), and domain name resolution (DNS), enabling applications like browsers and email clients to interact with the network and with each other.

Application Logic Explained

- **Definition:** Application logic is essentially the set of rules and syntax that govern how an application communicates with the network and other applications. It defines the messages, their format, and the expected sequence of interactions.
- **Role in the OSI and TCP/IP models:** It operates at the highest layer of both the OSI Model and the TCP/IP Model. In the TCP/IP model, the Application Layer combines the functions of the Application, Presentation, and Session layers of the OSI model.
- **Examples of protocols:**
- **HTTP/HTTPS:** Used for accessing web pages.
- **FTP:** Used for file transfers.
- **SMTP, POP, IMAP:** Used for sending and receiving emails.
- **DNS:** Used to translate domain names into IP addresses.
- **MIME**: Used to transmit multimedia data in emails.

Thank You